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Explainable Home and Last-Mile Connectivity Diagnosis

V.Ashritha
vashritha25@gmail.com

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Problem Description

- Users experience slow internet speeds, high latency, and frequent connection drops.
- It is difficult to identify whether the problem is caused by the Wi-Fi router, home network, or ISP connection.
- Existing tools often show network measurements without clearly explaining the cause of the problem.
- Users need a simple way to diagnose connectivity issues and understand possible solutions.
- The project aims to analyze network performance and provide understandable explanations and troubleshooting suggestions.

Background

- Internet connectivity is essential for online education, remote work, communication, and entertainment.
- Network performance depends on factors such as bandwidth, latency, packet loss, Wi-Fi signal strength, and network congestion.
- Home internet problems may occur due to router issues, wireless interference, or ISP connectivity problems.
- Common speed-testing tools measure network performance but may not clearly explain the reasons behind poor connectivity.
- An explainable diagnosis system can analyze network measurements, identify possible causes, and suggest suitable troubleshooting steps.

Literature & RFC Review

- Internet Performance Measurement: Existing tools such as Ookla Speedtest measure download speed, upload speed, and latency.
- Network Diagnostic Tools: Tools such as Ping and Traceroute help identify delays and connectivity problems.
- Wi-Fi Standards: IEEE 802.11 defines standards for wireless local area networks (Wi-Fi).
- Internet Standards (RFCs): Relevant RFCs define network protocols and diagnostic methods, such as ICMP for reporting network errors and supporting connectivity diagnostics.
- Research Gap: Many tools provide network measurements, but users may still need clearer explanations of possible causes and practical troubleshooting suggestions.

Proposed Solution

- Develop a system to measure internet performance using download speed, upload speed, latency, and packet loss.
- Analyze network measurements to identify possible connectivity issues.
- Distinguish between possible Wi-Fi problems, home network issues, and ISP-side connectivity problems where the available measurements allow.
- Provide simple explanations of detected issues instead of displaying only technical numbers.
- Suggest suitable troubleshooting steps based on the diagnosis.
- Display the results through a simple, user-friendly dashboard.

Optimization Proposed

Before Optimization	After Optimization
Users see only basic network measurements.	Users receive measurements with clear explanations.
Identifying the cause of slow internet is difficult.	The system identifies possible causes of connectivity issues.
Users may not know how to fix network problems.	The system suggests suitable troubleshooting steps.
Network issues are checked manually using different tools.	Network measurements are combined in one dashboard.
No clear summary of connection quality.	A simple report summarizes network performance.

Solution Architecture & Design

Visual: Flowchart or diagram showing the optimization process.

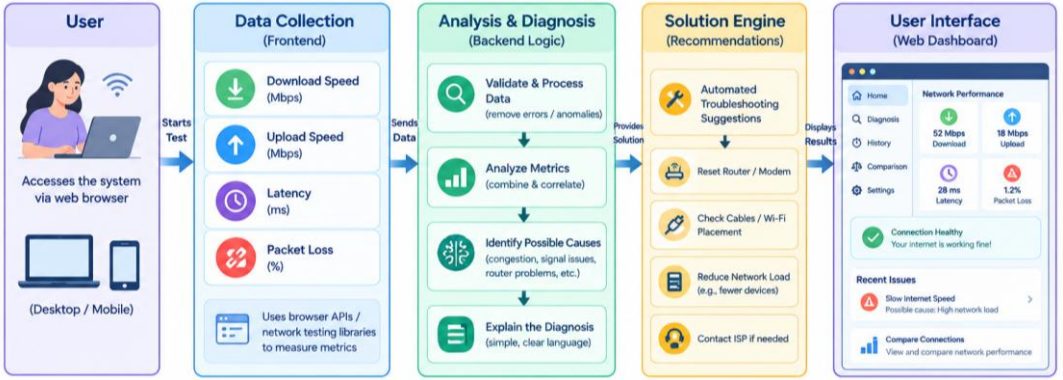


Figure: System Architecture UML Diagram

Data Flow Diagram

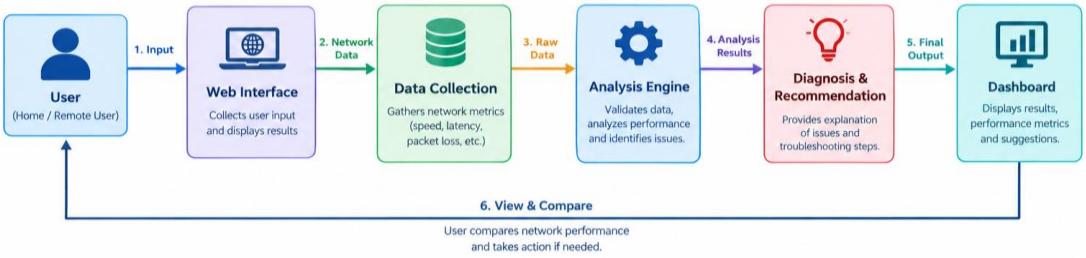


Figure: Data Flow Diagram (DFD)

Deployment Timeline

Milestones & Methodology
Phase 1 – Planning: Define requirements and select technologies.
Phase 2 – Development: Build the web interface and network analysis modules.
Phase 3 – Integration: Integrate diagnosis logic and troubleshooting recommendations.
Phase 4 – Testing: Validate accuracy, performance, and usability.
Phase 5 – Deployment: Host the application and make it accessible to users.
Phase 6 – Monitoring: Collect feedback, fix issues, and improve performance.

Methodology: Agile development with iterative implementation, testing, and continuous improvement.

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Thank You for Your Attention!